

A morphologically preserved multi-resolution TIN surface modeling and visualization method for virtual globes



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ABSTRACT

Virtual globes play an important role in representing three-dimensional models of the Earth. To extend the functioning of a virtual globe beyond that of a “geobrowser”, the accuracy of the geospatial data in the processing and representation should be of special concern for the scientific analysis and evaluation. In this study, we propose a method for the processing of large-scale terrain data for virtual globe visualization and analysis. The proposed method aims to construct a morphologically preserved multi-resolution triangulated irregular network (TIN) pyramid for virtual globes to accurately represent the landscape surface and simultaneously satisfy the demands of applications at different scales. By introducing cartographic principles, the TIN model in each layer is controlled with a data quality standard to formulize its level of detail generation. A point-additive algorithm is used to iteratively construct the multi-resolution TIN pyramid. The extracted landscape features are also incorporated to constrain the TIN structure, thus preserving the basic morphological shapes of the terrain surface at different levels. During the iterative construction process, the TIN in each layer is seamlessly partitioned based on a virtual node structure, and tiled with a global quadtree structure. Finally, an adaptive tessellation approach is adopted to eliminate terrain cracks in the real-time out-of-core spherical terrain rendering. The experiments undertaken in this study confirmed that the proposed method performs well in multi-resolution terrain representation, and produces high-quality underlying data that satisfy the demands of scientific analysis and evaluation.

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1. Introduction

Virtual globes provide a multi-resolution and intuitive three-dimensional (3D) digital representation of the planet (Goodchild, 2008). As a result of their continuous development, virtual globes can now incorporate not only global geospatial information, but also local research findings and analysis results in an interactive 3D environment. They offer a number of advantages over the traditional geographic information systems (GISs) when dealing with global environmental change issues (Yu and Gong, 2012). However, the power of the existing virtual globes is still restricted to functioning as a “geobrowser”, which is insufficient for scientific analysis and assessment (Craglia et al., 2012; Yu and Gong, 2012).

There are many reasons for this, but one of the most significant reasons is that their designers usually sacrifice accuracy for computational efficiency in the large-scale data management and visualization. However, to serve as scientific instruments, the scientific principles used to control the data quality in virtual globes should be of special concern (Goodchild et al., 2012).

A virtual globe is able to collect, manage, and display heterogeneous geospatial information from local to global scales, among which terrain data are the most fundamental type. Multi-resolution terrain modeling and visualization in a virtual globe forms the basis for integrating other types of geospatial data to render the planet, analyze the Earth-related phenomena, and display the analysis results. The modeling and representation of topography also enables a better understanding of the Earth surface processes which contribute to shaping the landscape. Numerous methods have been proposed for large-scale terrain processing and interactive rendering (Pajarola and Gobbetti, 2007), which can

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be categorized into two main groups: the grid-based methods and the triangulated irregular network (TIN)-based methods.

The grid-based methods use a regular grid model for the real-time terrain representation. Many algorithms have been developed based on grid-based terrain models, including the quadtree algorithm (Pajarola, 1998; Lee and Samet, 2000), the continuous level of detail (CLOD) algorithm (Lindstrom et al., 1996), the stateless, one-pass adaptive refinement (SOAR) algorithm (Lindstrom and Pascucci, 2001), and the real-time optimally adapting meshes (ROAM) algorithm (Duchaineau et al., 1997). The grid-based methods use meshes with a semi-regular or regular connectivity, together with fixed subdivision rules (Pajarola and Gobbetti, 2007), and they are suitable for the implementation of out-of-core terrain rendering based on tiled blocks. Therefore, most of the current virtual globes utilize a grid-based framework. However, the grid-based rendering models do not support highly detailed topography, such as LiDAR DEMs or point clouds (Isenburg and Shewchuk, 2009; Nebiker et al., 2010). As a result, most of the current virtual globe systems cannot be applied to high-resolution applications. More importantly, the grid geometry has difficulty in adapting to terrain irregularity, which results in large geometric redundancy and terrain expression distortion (Paredes et al., 2015). The distorted terrain representation can swear the analysis results displayed on it, and thus affect the user's understanding.

In addition to the multi-resolution rendering algorithms, another significant aspect devaluing the ability of the current virtual globe systems is the data processing accuracy. Given the huge amount of global terrain data, virtual globes need to generate the multi-resolution model in advance; for example, by creating pyramid layers or the level of detail (LOD). Image processing algorithms, such as resampling methods and wavelet methods, are usually used as the core techniques for multi-resolution pyramid generation in grid-based virtual globes. However, these methods usually ignore the natural characteristics of the landscape, which leads to serious topographic information loss during resampling (Ai and Li, 2010; Yue et al., 2015; Zheng et al., 2016). In such a case, the accuracy loss in the multi-resolution modeling results causes uncertainty in the following scientific analysis, and thus limits the application of virtual globes in scientific fields.

Another common approach to multi-resolution terrain modeling and visualization is the use of TIN-based methods, which are theoretically capable of producing a terrain representation of minimal complexity. Early examples of TIN-based methods can be found in the works of Hoppe (1997), Pajarola et al. (2002) and Ribelles et al. (2002). However, these methods were not all successfully applied to spherical terrain rendering in virtual globes, due to the high cost of maintaining the mesh connectivity, hierarchies, and dependencies. Researchers then strived to achieve a less costly implementation by imposing constraints on the mesh connectivity and update operations (Cignoni et al., 2003; DuYing, 2005; Gobbetti et al., 2006). Nevertheless, these methods are not fully TIN-based, which means that they are not able to blend irregular data due to the restrictions on the model structures. More recently, Christen and Nebiker (2011) presented a TIN-based approach for processing large-scale LiDAR elevation points for virtual globe visualization. However, the multi-resolution modeling strategy only provides real-time terrain rendering, while rigorous quality control of the generated multi-scale TIN data is omitted. Moreover, a solution to terrain crack elimination at run-time is not considered.

Based on these facts, we propose a TIN-based method for the processing of large-scale terrain data for virtual globe visualization and analysis. The point-additive algorithm is used as the base to construct the multi-resolution TIN pyramid from the input elevation data. Moreover, in the process of multi-resolution terrain

pyramid generation, cartographic principles are employed to connect each layer TIN with a scalable variable so that the TINs are controlled with a data quality standard. The drainage features are then extracted and embedded into the TIN model in each layer to preserve the basic shapes of the terrain surface at different levels. To enable efficient storage and organization, the generated TIN model in each layer is seamlessly partitioned based on a virtual node (VN) structure and tiled with a global quadtree structure. The continuous out-of-core spherical terrain rendering is implemented in a virtual globe system known as VirtualWorld v1.0. An adaptive tessellation approach is finally used to eliminate terrain cracks in the real-time visualization.

The rest of this paper is organized as follows. The multi-resolution TIN surface modeling is detailed in Section 2, while the multi-resolution TIN surface visualization is presented in Section 3. The experimental results and discussion are given in Section 4 and 5, respectively. Finally, the conclusions are drawn in Section 6.

2. Morphological modeling of the multi-resolution TIN pyramid

2.1. LOD control considering cartographic principles

The scale effect of data is considered to be one of the most fundamental issues in the geosciences (Quattrochi and Goodchild, 1997). Thus, the fact that the data quality requirement is dependent upon the data scale needs to be considered in multi-scale applications. In virtual globe realization, LOD algorithms are essential in multi-resolution modeling, as they are used to control the spatial scale in the different pyramid layers (total amount of geometry in a data tile) (Levenberg, 2002). For a traditional raster pyramid, the LOD for each layer is determined by the spatial resolution of the raster data (grid size). Differing from raster data, the resolution of TINs adapts to the terrain, rather than being fixed (Todd and Toma, 2006). Therefore, the LOD for TINs is usually an ambiguous term with respect to the spatial scale. In this paper, we employ a scale-dependent error tolerance to determine the specific LOD (degree of simplification) for each layer in the TIN pyramid construction. As the data quality criteria are dependent upon the spatial scale, the proposed LOD algorithm considering cartographic principles can help to control the quality of the processed data, to satisfy the demands of multi-scale applications. In the field of cartographic generalization, the maximum elevation error $E_{\max}$ and the root-mean-square error (RMSE) have been widely used as the data quality standards for topographical mapping (Zhou and Chen, 2011). For mapping at different scales, the permitted quality indices (e.g. RMSE and $E_{\max}$) for the corresponding spatial resolution are usually specified by the surveying and mapping agencies (State Bureau of Surveying and Mapping, 2001). However, it has been found that the two variables of RMSE and $E_{\max}$ are highly correlated (Zhou and Chen, 2011). Therefore, in our study, only $E_{\max}$ is used as the scale variable to control the LOD generation when constructing the TIN pyramid. According to cartographic principles, the relationship between the grid resolution D_r (local reference system, unit in meters) and the permitted $E_{\max}$ can be given as follows (Chen and Zhou, 2013):

$$\frac{E_{\max}}{D_r} = \frac{4}{5} \quad (1)$$

In virtual globe systems, once the spherical resolution D (WGS84, unit in degrees) for a given pyramid layer is determined, we can obtain the local spatial resolution D_r by coordinate transformation. The LOD is determined by the permitted $E_{\max}$, which can then be calculated based on (1).

The specific multi-resolution TIN pyramid structure can be constructed as follows. In a typical virtual globe tile system, the top level in the pyramid usually divides the whole of the Earth's surface into two basic tiles, and tiles for the other levels are generated based on a recursive quadtree subdivision of the top basic tiles. Given the input dataset with a spherical resolution D_c (along the longitudinal direction), the number of pyramid layers is determined by the following constraint:

$$\frac{D_{top}}{2^n} < D_c < \frac{D_{top}}{2^{n+1}} \quad (2)$$

where D_{top} is the resolution of the top layer, and n is the level of the bottom layer L_{bottom} . To match the spatial scale of the image data, all the tiles in the terrain pyramid should have the same size of 256×256 , and thus $D_{top} = 360/(nRootCountX \times 256)$, where $nRootCountX$ is the number of tiles in the top layer along the longitudinal direction. Based on (2), the level of the bottom layer can be calculated as:

$$n = \text{floor}(\log_2(D_{top}/D_c)) \quad (3)$$

where $\text{floor}(\cdot)$ rounds down to the nearest integer. Moreover, considering the relationship between D_c and the simulated resolution of the bottom layer, the number of layers L in the pyramid is adjusted as follows:

$$L = \begin{cases} n & \left| D_c - \frac{D_{top}}{2^n} \right| \leq \left| D_c - \frac{D_{top}}{2^{n+1}} \right| \\ n + 1 & \left| D_c - \frac{D_{top}}{2^n} \right| > \left| D_c - \frac{D_{top}}{2^{n+1}} \right| \end{cases} \quad (4)$$

For a standard pyramid with n layers, the spherical resolution of L_{bottom} is $D_{top}/2^n$. Otherwise, there will be $n + 1$ layers in the multi-resolution pyramid, and D_c will be used as the spherical resolution of the bottom layer. For each layer, the LOD in the TIN pyramid can be determined by the scale-dependent error tolerance E_{max} according to (1).

2.2. Morphologically preserved multi-resolution TIN pyramid construction

Surface simplification using TINs to create an adaptive multi-resolution terrain representation has garnered a lot of attention (Garland, 1999; De Floriani et al., 2005; Chen and Zhou, 2013). Among the previous methods, the point-additive algorithm with a maximum z-tolerance measure (also referred to as the maximum z-tolerance algorithm) has shown its superiority, and it can reconstruct an approximate surface using the minimal number of points for a given error tolerance (Garland and Heckbert, 1997). Moreover, the method is easy to implement and efficient when using a greedy insertion approach (Chang, 2006). The point-additive algorithm features an iterative process starting with an initial triangulation to approximate the surface, and it then recursively inserts the points with the highest error into the current triangulation, until no points remain with an error higher than the given error tolerance. This method can be adapted to various input data types, and finally produces a hierarchical triangulation based on the iterative process. In our application, the pyramid layers from the top to the bottom are derived from different hierarchical levels of the triangulation, based on the scale-dependent error tolerance E_{max} .

However, the traditional point-additive algorithm only derives isolated points to form the TIN during the iterations, which cannot represent the continuous linear patterns of a natural landscape. As a result, the morphological information is lost in the surface generalization, which cause uncertainties in the further analysis. It is therefore necessary to incorporate feature constraints to preserve the significant topographic characteristics in multi-resolution

modeling (Passalacqua et al., 2010). In the field of cartographic generalization, drainage lines are often employed as constraints to preserve the structural information and the basic morphological shapes of landscapes at different resolutions (Ai and Li, 2010; Zhou and Chen, 2011). The previous methods mainly work on grid DEMs and use the simple “D8” flow routine algorithm to extract the drainage lines, which can cause serious topological errors, especially on complex terrain (Jasiewicz and Metz, 2011). In this study, a hybrid method developed in our previous work is applied for high-quality drainage network extraction (Zheng et al., 2015). This method incorporates discrete curve evolution (DCE) to eliminate the noise influence, and utilizes a skeleton construction technique to distill accurate channel paths over complex terrain. It can derive topologically correct drainage lines from different types of data, even in cases with significant terrain noise and irregularities. More details about the drainage extraction method can be found in our previous work (Zheng et al., 2015).

The multi-resolution TIN pyramid construction process is illustrated in Fig. 1. Given an input elevation dataset, the number of layers and the permitted error tolerance for the LOD control in the different pyramid layers can be calculated as described in Section 2.1. The point-additive algorithm is performed to iteratively construct the TIN pyramid by a top-down approach. During the iterative process, when the error between the candidate point to be inserted and the current TIN surface is less than the error tolerance of a certain layer, the TIN hierarchy level is derived for that layer. For example, if the errors between all the points and the TIN surface are less than ε_m (i.e., the permitted E_{max} for layer m , $m \in [0, n]$), the TIN for layer m is derived. The extracted drainage lines are then generalized using the Douglas-Peucker algorithm to match the scale of that layer, and are embedded into the derived TIN structure.

3. Visualization

3.1. TIN tile storage with the virtual node (VN) structure

Efficient storage and access are important when processing large-scale data, especially when modeling and rendering terrain data in a global context (Kolar, 2004). In this section, we provide the details of the VN structure and the strategy of the virtual partitioning for efficient TIN tile storage. During the iterative process of multi-resolution TIN modeling, virtual partitioning is performed on each pyramid layer after the TIN structure is constructed, as described in Section 2.2. Fig. 2 indicates the process of virtual partitioning performed on a given level of triangulation. In Fig. 2(b), the dotted lines in red indicate the tile edges resulting from the global quadtree subdivision of the Earth's surface, while the red points are the inserted tile corner points. After inserting the corner points, the original triangles containing the corner points are partitioned as shown with the green lines in Fig. 2(b). As the corner points are part of the triangulation (vertices), it is easy to walk from a corner point to another corner point through the Delaunay edges, and to calculate the other border points at the points where the tile edges intersect the Delaunay edges. A specific example is given in Fig. 2(c). All the border points are treated as the constraints in the triangulation, and their elevations are calculated by undertaking a linear interpolation at the corresponding position in the triangle containing the point.

The virtual partitioning is based on the fact that the tile border points are artificially appended virtual vertices, rather than the actual vertices necessary for triangulating the terrain surface. The virtual vertices are designed to generate a rectangular tile border for TIN data storage and access. It is also worth noting that they are independently generated for each TIN layer. The process of

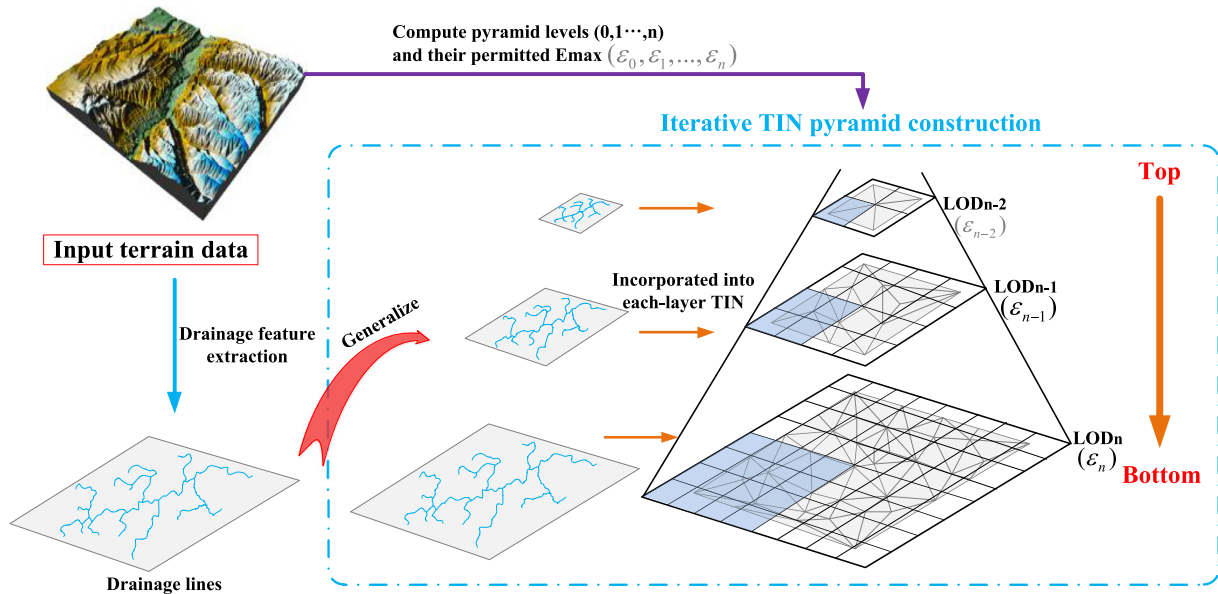


Fig. 1. The multi-resolution TIN pyramid construction process.

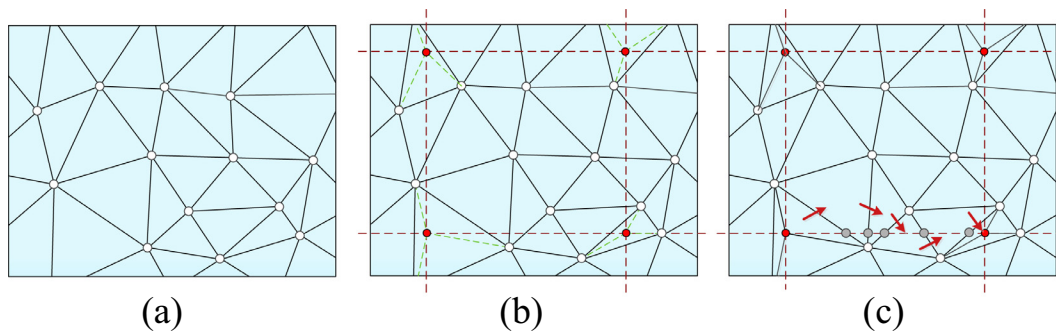


Fig. 2. Virtual partitioning. (a) Original triangulation. (b) Insert tile corner points. (c) Quickly walk from a corner point to another corner point through the Delaunay edges and calculate the other border points.

virtual partitioning helps blend the TIN models with the raster image tiles or integrate them with the grid mesh tiles. Furthermore, the rectangular tile border allows the large-scale triangulation to be divided into independent tile blocks, and thus the multi-resolution TIN pyramid can be stored in a distributed database and scheduled in a similar way to the raster tiles.

To enable the efficient storage and organization of the multi-resolution TIN pyramid, we define the structure of a TIN tile as shown in Fig. 3. Here, the triangles with one edge lying on a tile boundary are defined as virtual triangles, which are the pink

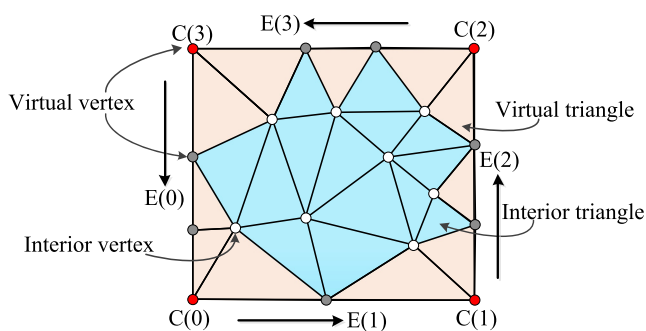


Fig. 3. Structural component definition of a TIN tile.

triangles in Fig. 3. The VN structure is then composed of virtual vertices and virtual triangles. Following the description of the VN structure, we denote the edges of a tile as $E(i)$ and its corners as $C(i)$, with $i = 0, \dots, 3$. By traversing the quadtree structure in a Morton order, tiles in the current layer are written into the database, based on the VN structure. Considering that the corner points and tile edge points are shared by the neighborhood tiles, they are stored in a corner structure and an edge structure, respectively, and are distinguished from the interior points in a TIN tile. The other information associated with a TIN tile, including the interior vertices, the list of virtual triangles, and the interior triangles, is stored in the tile body structure and can be indexed with a quad-tree key (i.e. “quadkey”). Based on the quadkey of the tile body structure, the data stored in the corner structure and the edge structure can build their own index keys so that they can be accessed individually. To reduce the redundancy of the storage structure, only the left bottom corner point and the two adjacent edges (i.e. $C(0)$, $E(0)$ and $E(1)$) are saved when writing out a tile, to ensure that the corner points and the tile edges are only stored once for the entire TIN layer. Moreover, the list of virtual triangles along the four edges $E(0), \dots, E(3)$ is stored in a sequential structure with a counterclockwise order. The sequential structure is designed for the implementation of the algorithm for the TIN terrain crack elimination, which is detailed in Section 3.2.

At run-time, meshes for the TIN tiles are assembled based on the points and triangles stored off-line. Since the triangles are defined by point indices, to guarantee that the points stored in the different data structures can be assembled in the correct order and correctly indexed, we associate a local index for all the points in a tile before writing out the triangles, as illustrated in Fig. 4. Moreover, the association of local indices for the border points follows a counterclockwise order, starting from corner $C(0)$, along edge $E(0)$, $C(1)$..., and ending with the last point on edge $E(3)$. The triangles are then written out based on the local indices of the vertices. Taking Fig. 4 as an example, when preparing the mesh for a tile at run-time, the points can be assembled as follows:

$$V_{list} = \{V_{interior}[v_0, v_1, v_2, v_3, v_4, v_5, v_6, v_7], C(0)[v_8], E(1)[v_9], \\ C(1)[v_{10}], E(2)[v_{11}, v_{12}], C(2)[v_{13}], E(3)[v_{14}, v_{15}], \\ C(3)[v_{16}], E(0)[v_{17}, v_{18}]\} \quad (5)$$

where the order is the same as the order of the local index association. Thus, the triangles can be correctly reconstructed based on the stored point indices.

3.2. Interactive spherical rendering of the multi-resolution TIN

The interactive spherical rendering of the multi-resolution TIN terrain is implemented in a virtual globe system named VirtualWorld v1.0, as shown in Fig. 5, which was developed by Wuhan University. The design principles of this virtual globe system are similar to those of the typical grid-based virtual globes; for example, the hierarchical quadtree-based multi-resolution data organization, the data scheduling, the LOD assignment of surface tiles, and the view-dependent rendering. In this study, the multi-resolution TIN models are tiled and stored with the quadtree structure. In this section, we describe the key techniques for the continuous terrain rendering implemented in VirtualWorld v1.0.

One of the main differences between the TIN-based and grid-based models lies in the terrain crack elimination algorithm. In virtual globe applications, one of the most common approaches is to use a curtaining method to build a vertical polygonal wall to hide the terrain cracks. This approach can avoid real-time retriangulation, as described in Christen and Nebiker (2011). However, the curtaining methods have problems dealing with large terrain gradients. Moreover, the terrain models with different LODs require matching triangulation to maintain the surface continuity in some applications (Agugiaro and Kolbe, 2012; Paredes et al., 2015). For the TIN models, terrain crack elimination is a relatively complex issue, which requires an adaptive tessellation algorithm to generate additional primitives in real time to join the meshes with different LODs. In this study, we present a proposal in which part of the tessellation information is precomputed and efficiently

encoded. At run-time, the precomputed information is adaptively decoded and a tessellation procedure is performed to stitch the meshes with different LODs. In the following descriptions, we assume that the adjacency relations between the neighboring tiles with different LODs have been previously determined.

3.2.1. Tessellation precomputation and codification

The process of tessellation precomputation and codification is illustrated with an example in Fig. 6, where two tiles corresponding to different LODs coexist in the same scene. We denote the tile with a higher LOD as the high-resolution (HR) tile, while the tile with a lower LOD is called the low-resolution (LR) tile. To achieve a smooth transition between the LR and HR meshes, the unmatched triangles on the border of the LR and HR tiles have to be locally tessellated, as indicated by the red lines in Fig. 6(a). The tessellation algorithm used in the proposed method eliminates the terrain cracks with a simple sequential connection operation, thus avoiding retriangulation within the TIN tile.

Next, we take Fig. 6(b) as an example to illustrate the process of codification for the tessellation information. If the projected locations of the two extra vertices v_1 and v_2 lie on the border edge C_1V_1 of T_1 , then the two vertices are added to the vertex list of T_1 , and T_1 is tessellated by connecting its interior vertex I_1 to the new list of its border vertices $\{C_1, v_1, v_2, V_1\}$ following a sequential order. In this way, a new locally matched triangulation can be obtained with the three generated sub-triangles, and the original virtual triangle T_1 is displaced. To guarantee the efficiency of the tessellation at run-time, we adopt a precomputation strategy. Traversing the virtual triangles and the corresponding border vertices of neighboring tiles, the intersection relations of the tile edges and the border vertices are examined. Based on the relations of the neighborhood tiles and the border vertices, the virtual triangles need to be tessellated, and the vertices for tessellation can be recorded. Since the multi-resolution TIN pyramid is constructed and stored off-line, it is possible to compute and encode the related information in advance in order to reduce the computational cost at run-time.

In our proposal, we include three fields for each virtual triangle to encode its tessellation information. The first field uses a two-value index indicating whether or not a virtual triangle needs to be tessellated. The second field encodes the number of extra vertices to be included in tessellating the virtual triangle. The last field indicates if the second vertex on the border edge of the original virtual triangle (e.g., v_1 or V_1 in Fig. 6(b)) overlaps with a border vertex of a neighboring tile.

The tessellation codification for the list of virtual triangles along the four tile edges is performed sequentially but independently. Thus, each tile edge is associated with a complete code string composed of the tessellation code for all the virtual triangles along that

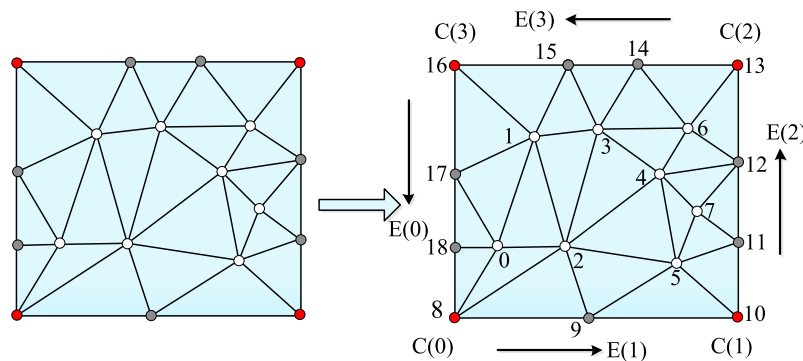


Fig. 4. The local index association for each point in a tile.

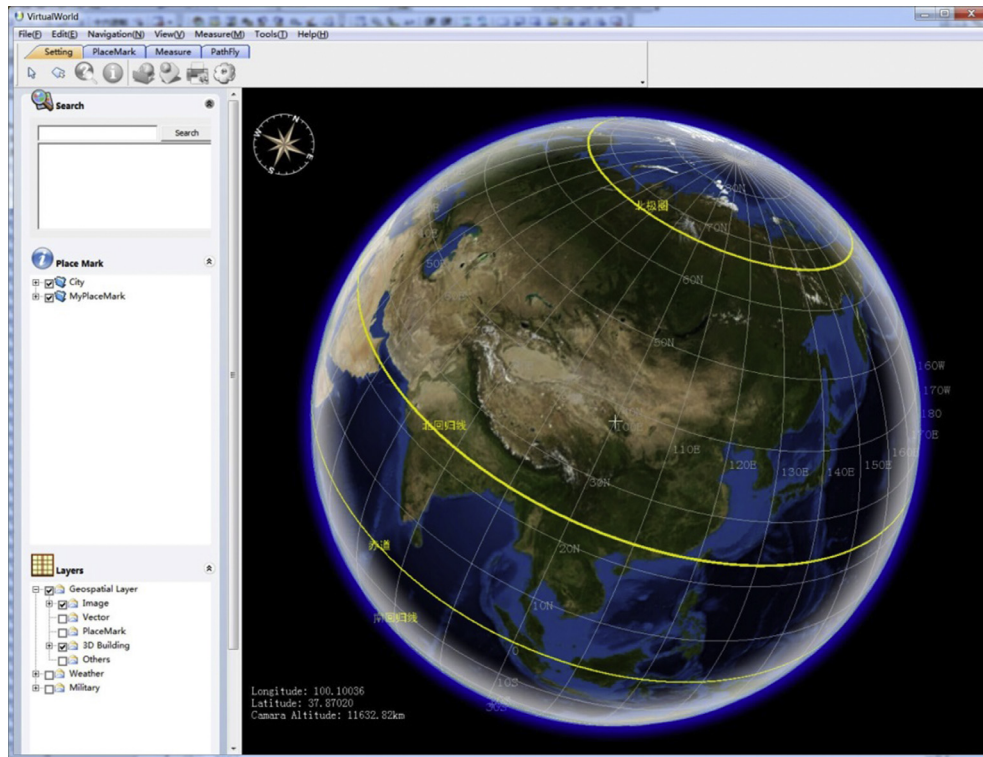


Fig. 5. The global view of the Earth's surface on the VirtualWorld v1.0 virtual globe system.

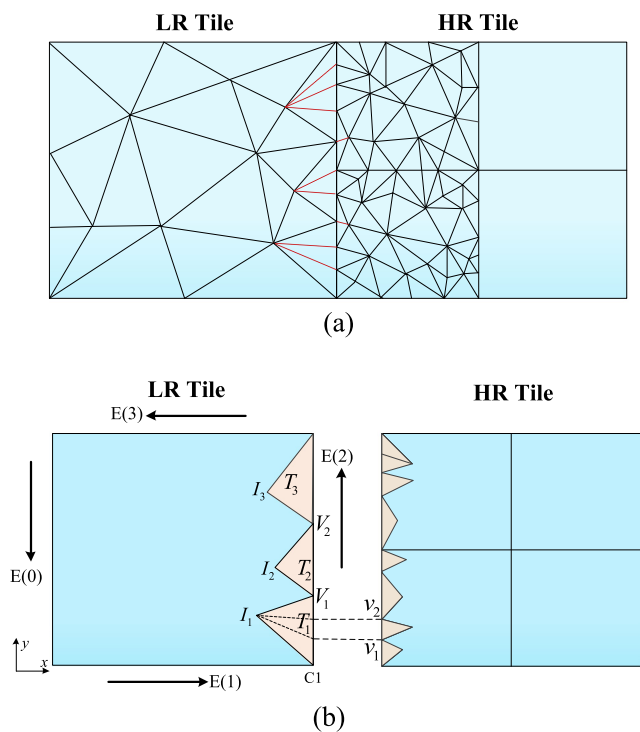


Fig. 6. Tessellation precomputation and codification. (a) Tiles with different LODs coexist in the same scene. (b) An example of computing the tessellation information for $E(2)$ of the LR tile.

edge. According to our proposal, the tessellation information associated with the virtual triangles along edge $E(2)$ in Fig. 6(b) can be given as:

$$\text{Tessellation}[E(2)] = \{T_1(1, 2, 0), T_2(1, 2, 0), T_3(1, 3, 0)\} \quad (6)$$

Based on this sequential structure, the code string associated with $E(2)$ can be expressed as 120120130 , which is stored in the tile body structure. Following the tessellation codification of the virtual triangles along the LR tile edges, the tessellation along the corresponding HR tile edges is also encoded. The whole procedure of tessellation codification is integrated into the multi-resolution TIN pyramid construction, along with the iterative TIN generation process. Therefore, it is easy to infer that each tile edge may actually associate with two code strings, where one of them corresponds to an LR tile and the other corresponds to an HR tile. To help the readers understand the proposed method, the main data storage structure of a TIN tile, including the tessellation information, is depicted in Fig. 7.

3.2.2. Mesh matching with tessellation decoding

Following the codification strategy, we now use Fig. 8 to briefly describe how the tessellation information is efficiently decoded for the view-dependent multi-resolution TIN mesh matching at run-time.

As presented in Section 3.2.1, the required code string associated with edge $E(2)$ is 120120130 , which is loaded from the tile body structure of the LR tile. According to our proposal of tessellation codification, each group of the three field codes in this code string constitutes the entire tessellation information of a virtual triangle. Therefore, for $T_1(120)$, the first field "1" indicates that T_1 needs to be tessellated. Given that the index information contained in the tile body structure is known, it is easy to retrieve T_1 from the triangle list based on the sequential organization structure of the virtual triangles. The second field indicates that there are two vertices lying on the border edge of the neighboring HR tile for T_1 . In Fig. 8, the two points (v_1, v_2) on edge $E(0)$ in the HR tile are retrieved, counted, and added to the vertex list of T_1 within the LR tile. The value of the third field here is 0, which means that vertex V_2 does not overlap with a border vertex of the neighboring HR tile. Therefore, the tessellation can be directly performed by a

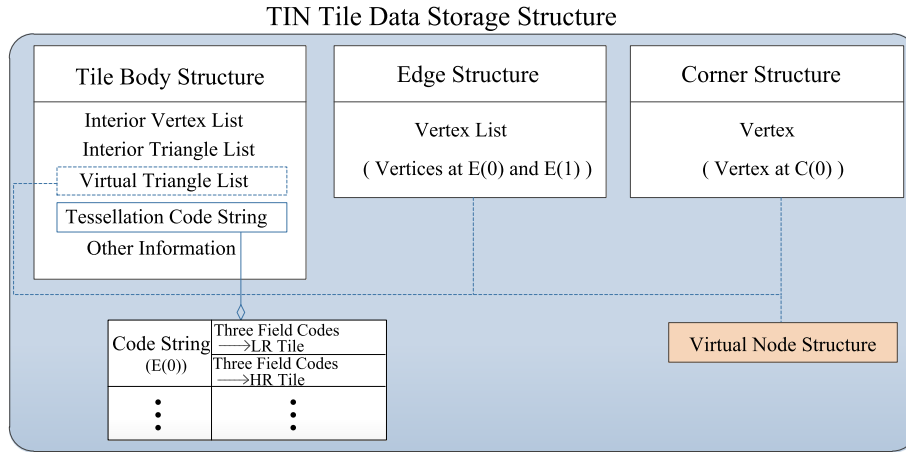


Fig. 7. The main data storage structure of a TIN tile.

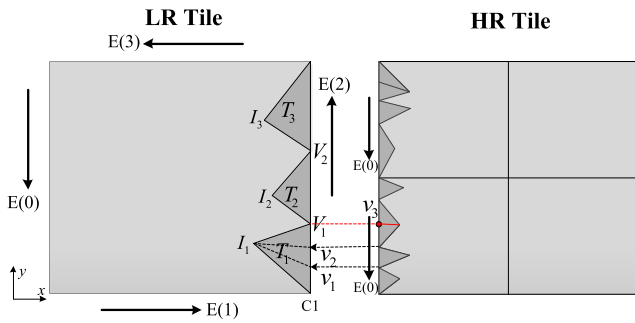


Fig. 8. Mesh matching with tessellation decoding.

consecutive connection of interior vertex I_1 with border vertices C_1 , v_1 , v_2 , and V_2 . Otherwise, if there has a vertex v_3 overlaps with V_2 , as indicated by the red point in Fig. 8, the elevation values of the two vertices should be carefully checked to avoid a potential terrain crack. In this case, vertex v_3 must also be counted before

processing the next virtual triangle, because the retrieval of the extra vertices for the tessellation of the subsequent virtual triangles is based on the number of previously retrieved vertices.

The proposed tessellation codification and decoding is efficient and compact. The key idea of this proposal is the use of the sequential structure, the index operations, and the pairing of the virtual triangles with the extra vertices for tessellation. Because there are no geometric operations to guarantee a correct local retriangulation at run-time, special attention should be paid when retrieving the extra vertices for the virtual triangle tessellation since they are strictly paired and count-based.

4. Results

For the purpose of comparison and evaluation, we conducted the first application on a grid DEM dataset. The data are from Wasatch Mountain near the city of Salt Lake, Utah, and the area is composed of 6402×8372 grid cells (roughly 64.0×83.7 km) with a 10-m spatial resolution, as shown in Fig. 9(a). The data

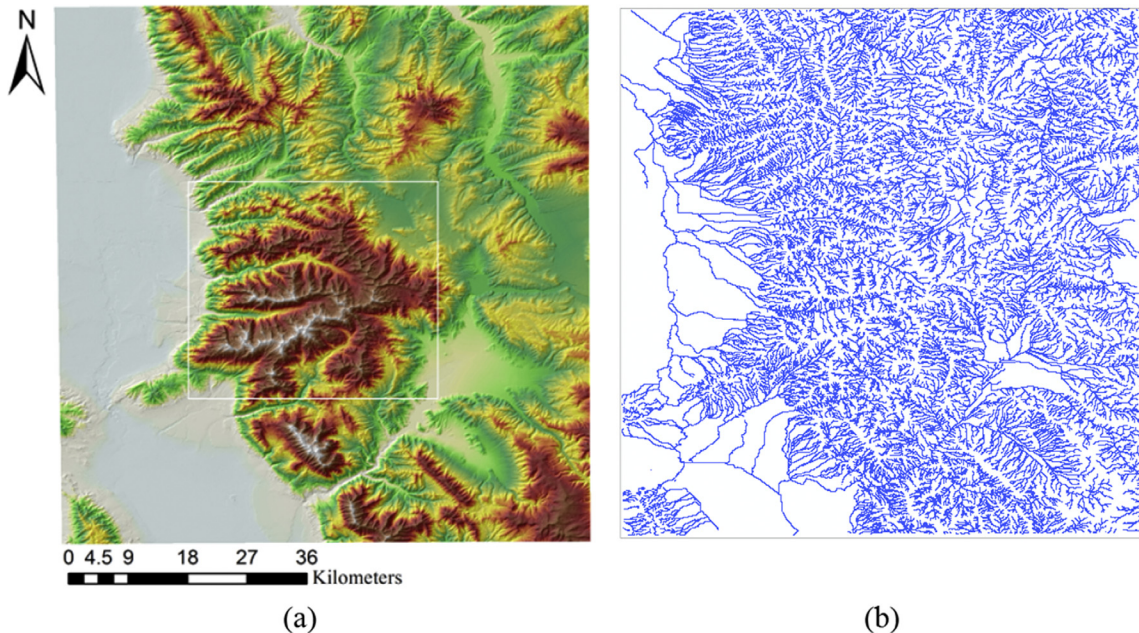


Fig. 9. The first dataset from Wasatch Mountain near the city of Salt Lake, Utah. (a) Original 10-m resolution DEM. The area marked with the white rectangle was chosen for the quantitative analysis in Section 5.1. (b) Topologically correct drainage features extracted from the DEM.

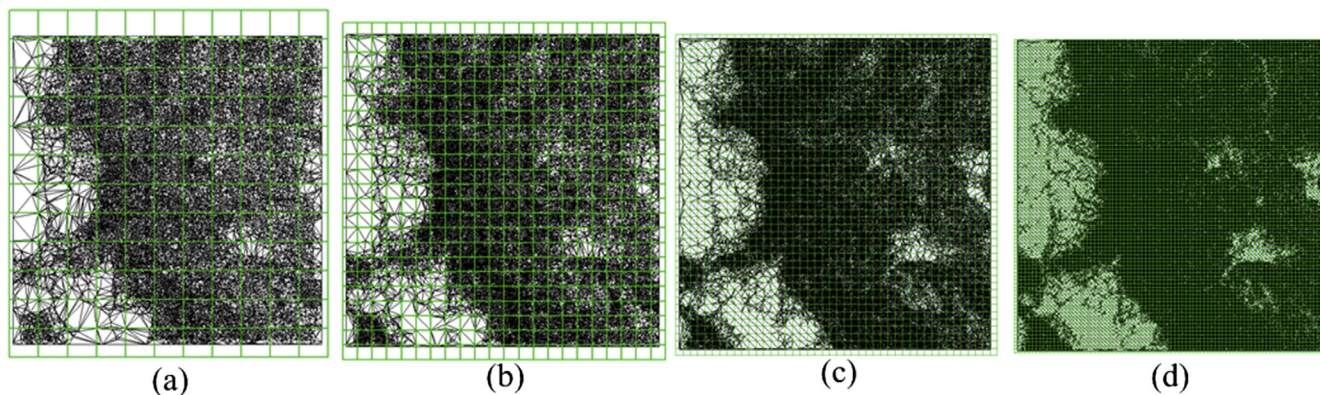


Fig. 10. The triangulation results and the corresponding global quadtree subdivision at the four finest levels: (a)–(d) represent the coarsest level to the finest level (bottom layer).

Table 1

The computational efficiency of the proposed method (unit: s).

Time	Level				
	1	2	3	4	5
TIN construction	25	7	8	17	21
Other operations	2	5	9	24	68

are available via the national elevation dataset (NED, <http://gis.utah.gov/data/>). The area shows significant terrain variations along the latitudinal direction, with a higher relief occupying the eastern

part and a flat topography in the western part. The study area is suitable for evaluating the performance of the proposed hybrid drainage feature extraction method, since many methods have difficulty in adaptively modeling a watershed in areas with compound terrain. Fig. 9(b) shows the topologically correct drainage networks delineated by the hybrid method (Zheng et al., 2015). The extracted drainage networks were then utilized in the following multi-resolution TIN modeling. For the convenience of presentation, we refer to the proposed morphologically preserved multi-resolution TIN modeling and visualization method as the M-TIN method hereafter.

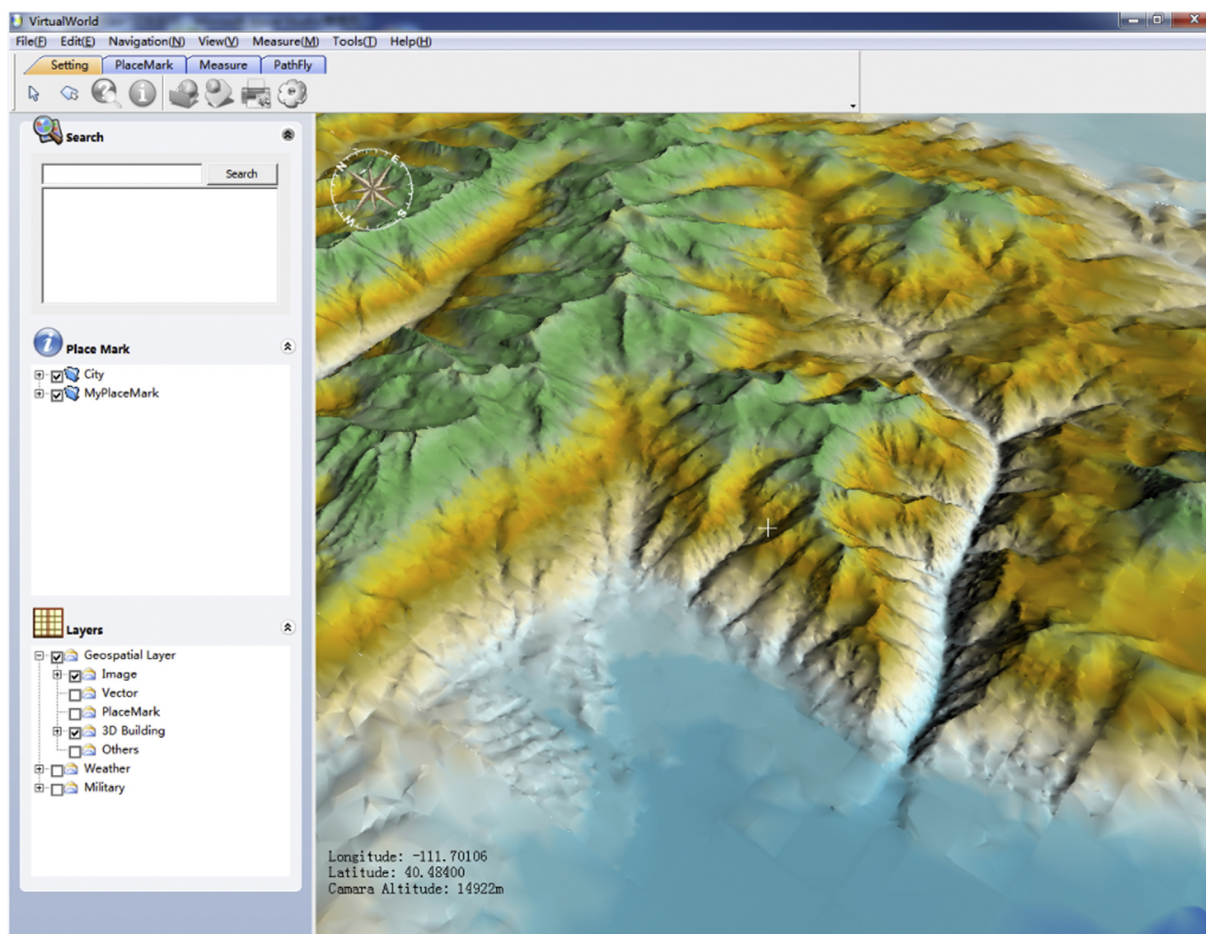


Fig. 11. Real-time rendering of the morphologically preserved multi-resolution TIN terrain, with hill-shade effect.

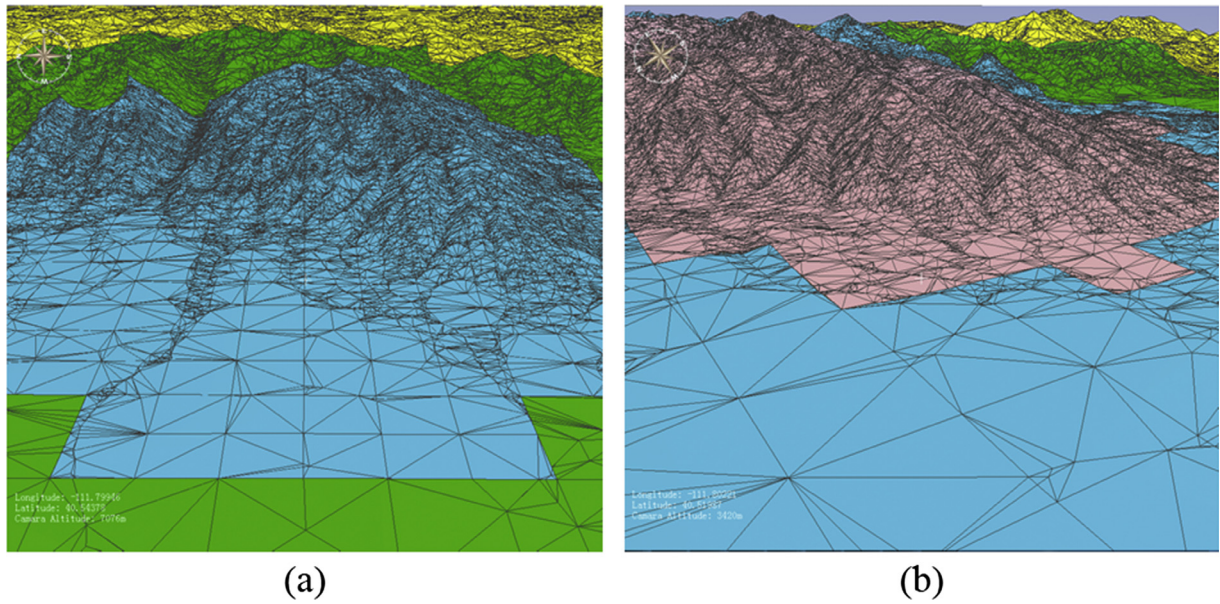


Fig. 12. Multi-resolution TIN terrain in wireframe mode to show the view-dependent LOD rendering.

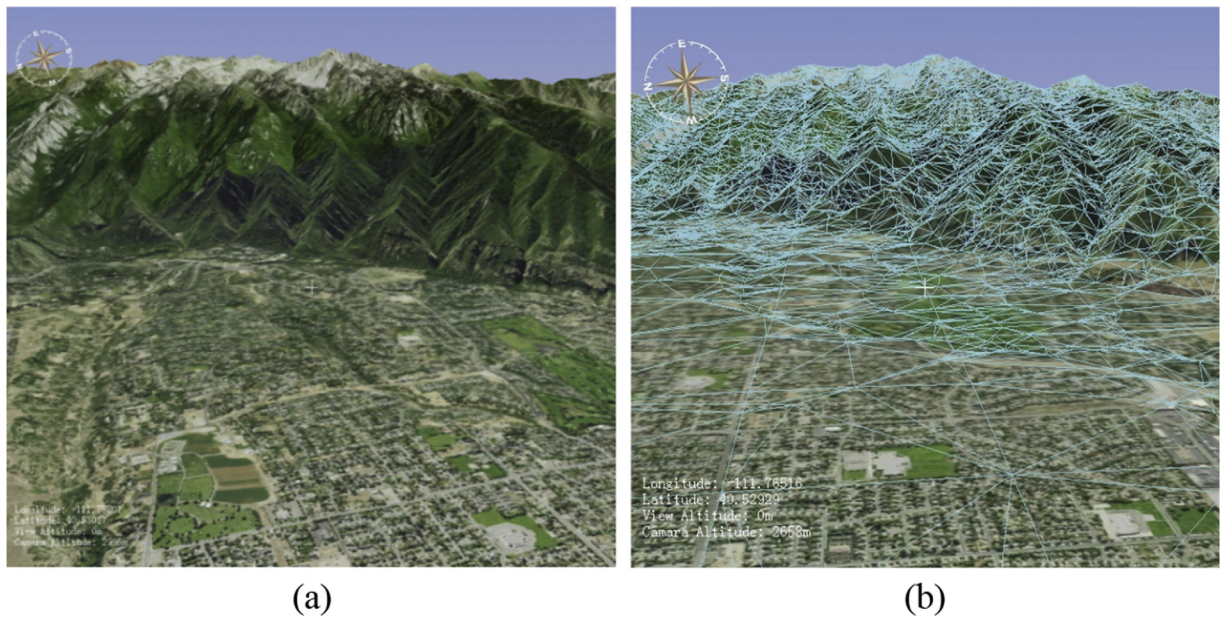


Fig. 13. Rendering the TIN with aerial imagery. (a) Textured terrain. (b) Elevation tiles in wireframe mode blended together with image tiles.

Table 2

The statistical results of the two methods.

Level	Method	Points ($\times 10^3$)	% (points)	Triangles ($\times 10^3$)	RMSE (m)	$E_{\max}$ (m)
1	M-TIN	21.2	0.2	60.1	31.51	128
	Resampling	599.35	6.25		32.23(64)	198
2	M-TIN	31.7	0.3	78.6	17.33	64
	Resampling	1198.7	12.5		15.18(32)	135(64)
3	M-TIN	65.8	0.7	142.4	8.99	32
	Resampling	2397.4	25		6.87(16)	97(32)
4	M-TIN	151.9	1.6	309.7	4.76	16
	Resampling	4794.8	50		2.74(8)	49(16)
5	M-TIN	423.9	4.4	847.1	2.48(4)	8(8)
	Resampling	9589.6	–		–	–

Before constructing the multi-resolution TIN pyramid for the test data, the local elevation dataset was first transformed from the source spatial reference system into a global reference system. Following the instructions in Section 2.1, the number of pyramid layers was set as five in the M-TIN modeling process, and the error tolerance $E_{\max}$ was then initialized for each pyramid layer. Fig. 10 shows the triangulation results at the four levels, where the first level is not given. The green cells in each of the sub-figures indicate the corresponding spatial subdivision at that level according to the global quadtree hierarchy.

The computational performance of the proposed M-TIN method with the test DEM was also evaluated. The main procedures include TIN construction and other operations, including the drainage line insertion, the TIN partitioning, and the tessellation codification. The program was written in C++ code, and the experimental computer was configured with an Intel(R) Core(TM) i3-2100 3.1 GHz CPU, 8 GB RAM, and a 64-bit Microsoft Windows 7 operating system. The results are shown in Table 1.

With the M-TIN pyramid pre-prepared and stored off-line in the database, the continuous LOD rendering of the M-TIN terrain was

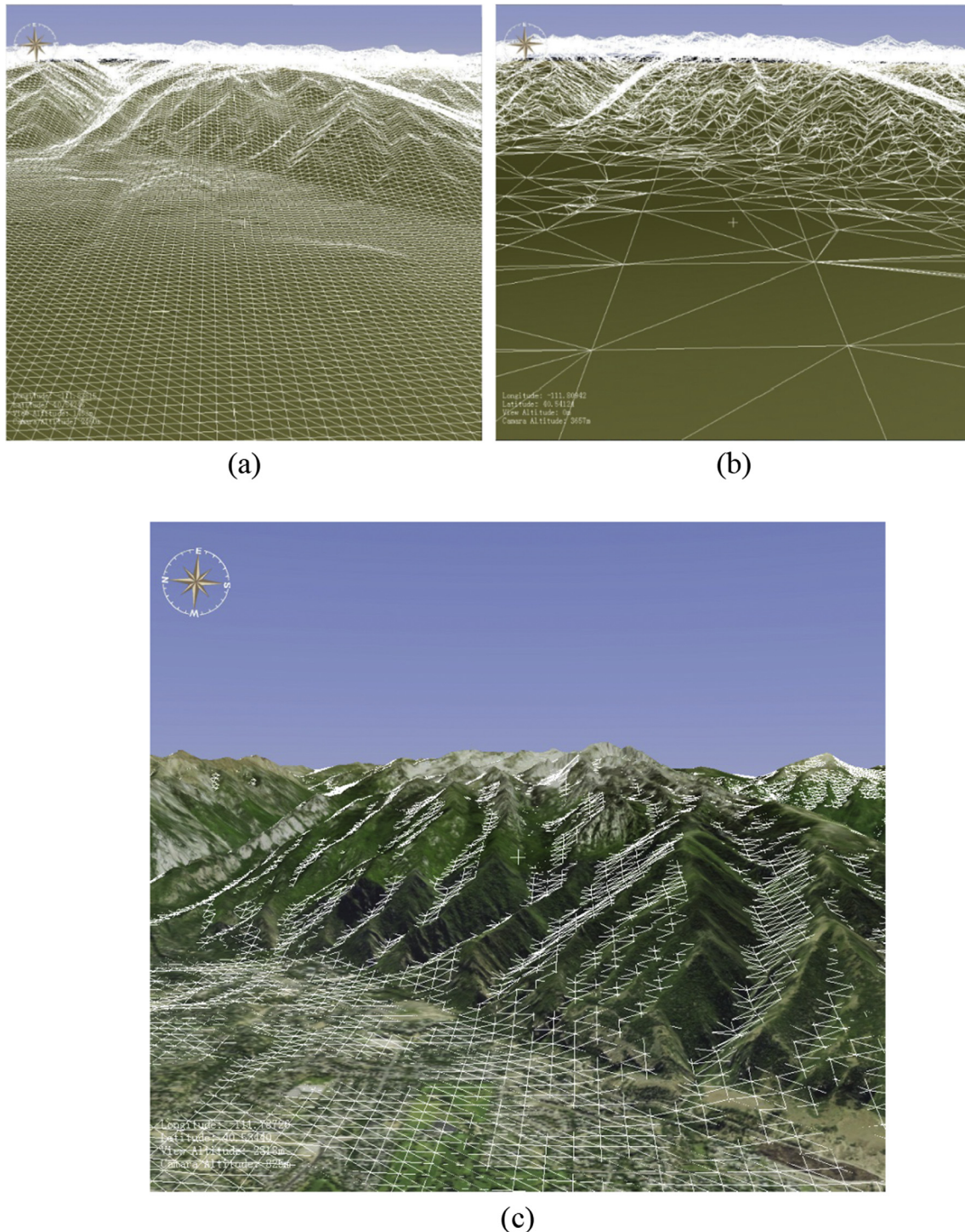


Fig. 14. Terrain representation with the grid-based approach (a) and the TIN-based approach (b).

implemented in the VirtualWorld v1.0 virtual globe system. Fig. 11 shows the 3D representation of the M-TIN surface, with the shaded effect in the Wasatch area implemented in the virtual globe system. In Fig. 11, the mountainous areas with sharp relief indicate that the terrain morphology is effectively preserved by the use of the proposed M-TIN method. Fig. 12 visualizes the wireframe of the multi-resolution TIN surface, where the data blocks displayed at different resolutions are rendered with different colors. The experimental performance demonstrates that the proposed method permits a run-time scheme for temporally smooth, view-dependent LOD control on arbitrary meshes.

The multi-resolution TIN tiles can also be easily integrated with the image tiles to generate textured terrain, as shown in Fig. 13. The image data are tiled using the same global quadtree structure as the elevation data. In Fig. 13(b), elevation tiles in wireframe mode are blended together with image tiles to show the consis-

tency between the geometric features and textural features in representing the surface morphology.

5. Discussion

In this study, we have explored the M-TIN method for the processing of large-scale regular or irregular elevation datasets for virtual globe visualization and analysis, which can overcome some of the limitations of the current grid-based virtual globes. A further evaluation with both statistical and visual comparisons follows.

5.1. The effectiveness of the M-TIN method for data processing

In order to evaluate the effectiveness of the proposed M-TIN method for data processing, the bilinear resampling method, which is commonly used for data processing in grid-based virtual globes,

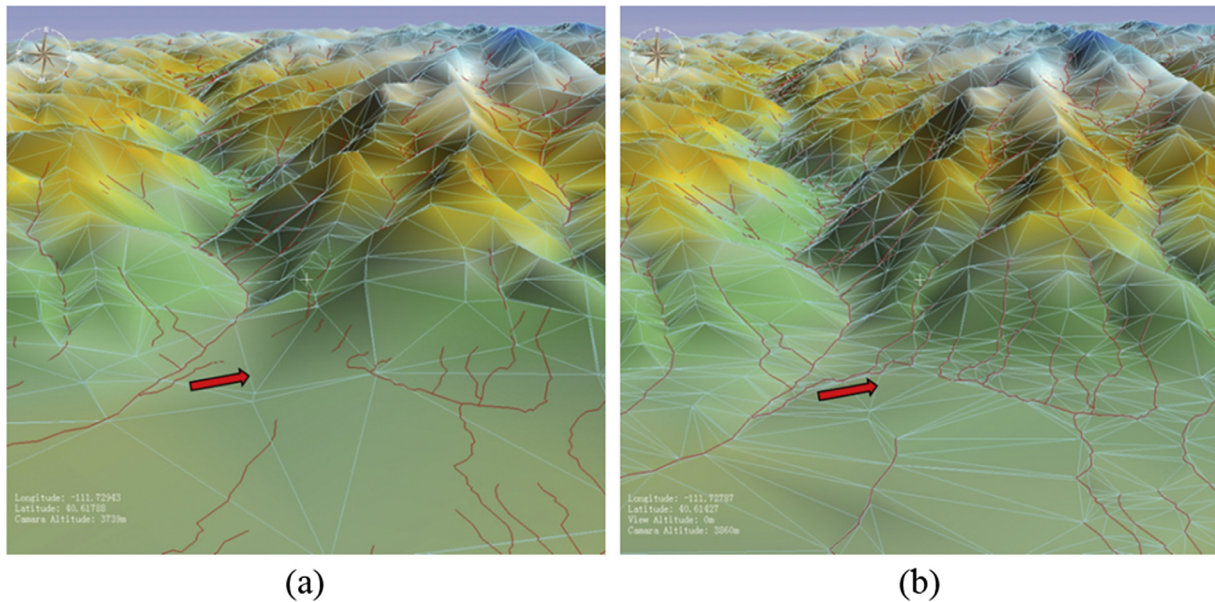


Fig. 15. Comparison between terrain representation with and without the drainage constraint, where the red lines are the stream lines. (a) Without the drainage constraint. (b) With the drainage constraint. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

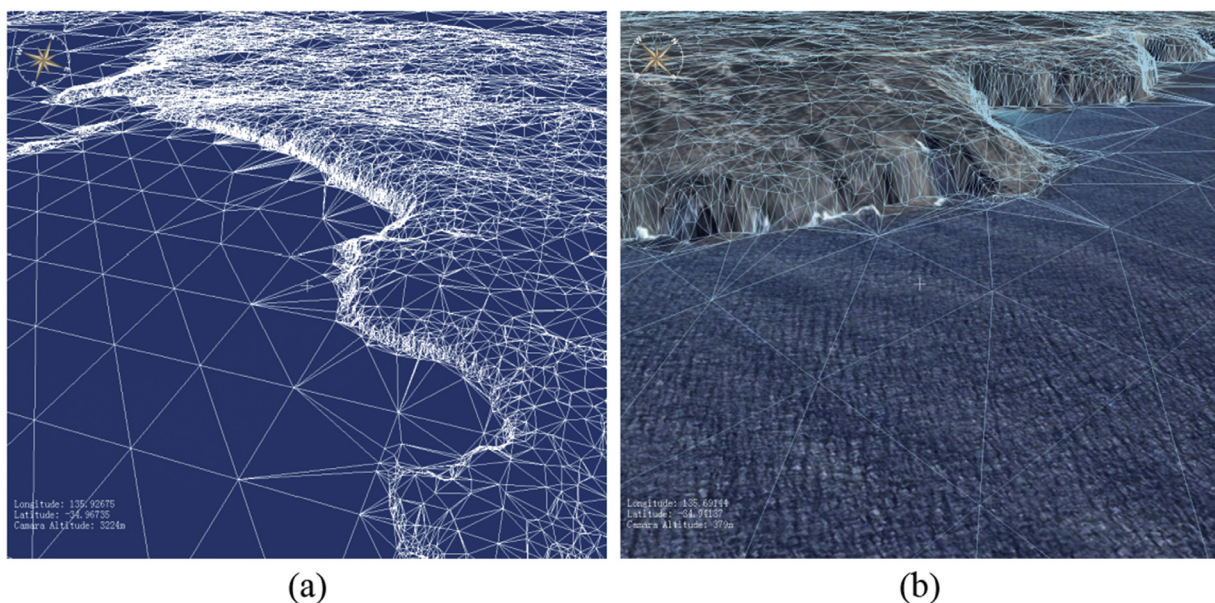


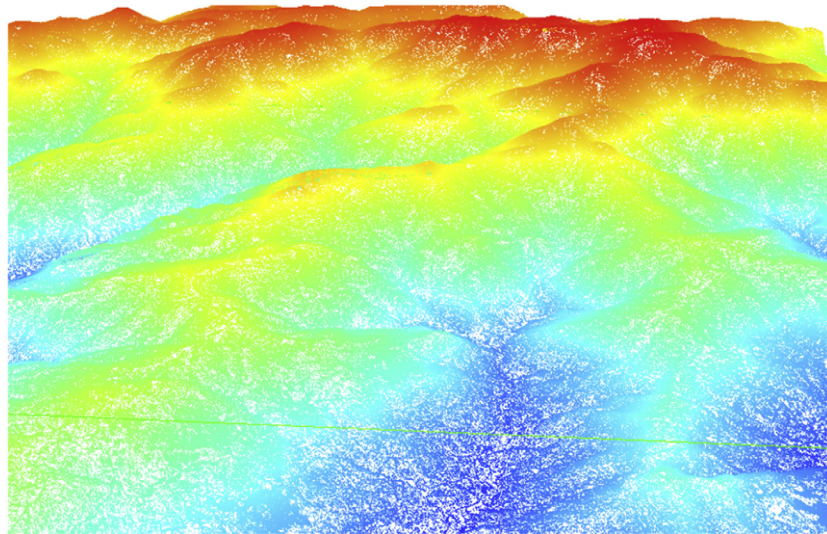
Fig. 16. The application of the proposed M-TIN approach on high-contrast land-sea topography. (a) TIN wireframe. (b) Blended with image data.

was chosen for the comparison. Considering that the whole Wasatch area, as shown in Fig. 9, was too large for the comparison, only the area marked with a white rectangle was chosen. To allow a fair comparison, a five-level raster pyramid for the DEM was constructed by the bilinear resampling method. In each layer, the raster resolution matched the scale of the TIN model (as presented in Section 4) at the same level. The statistical results of the two methods are compared in Table 2. Firstly, the number of points required to construct each layer in the multi-resolution pyramid is shown. Moreover, the generalized TIN and raster surface in each level were resampled to the original DEM to derive the statistics, where RMSE and $E_{\max}$ were employed as the quantitative measurements. In this way, we can give a quantitative comparison between the efficiency and accuracy of the different methods for multi-resolution modeling. Moreover, the percentage of points used for the multi-resolution modeling compared with the original grid data is given to indicate the degree of data simplification in each level.

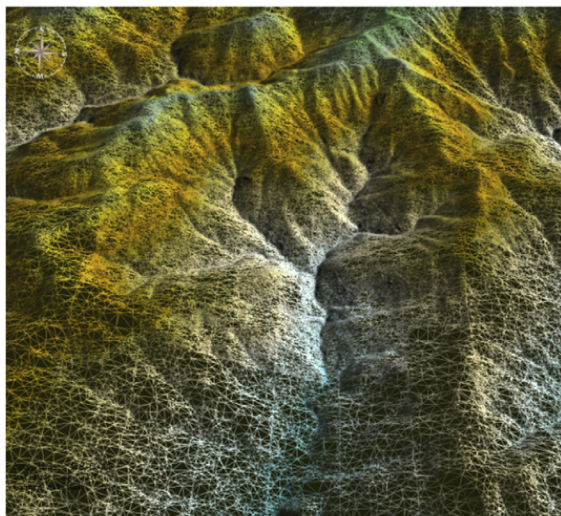
The comparison results show that the M-TIN approach can efficiently simplify the original DEM using a very limited number of points, gaining an RMSE that is close to that obtained by the resam-

pling method. The RMSEs of both methods are well below the accuracy tolerance permitted by the cartographic standard (as listed in brackets). However, the resampling method fails to deliver results that meet the mapping standard in the measurement of $E_{\max}$. In contrast, by employing the cartographic principles, the M-TIN method performs better at controlling the measurement of $E_{\max}$.

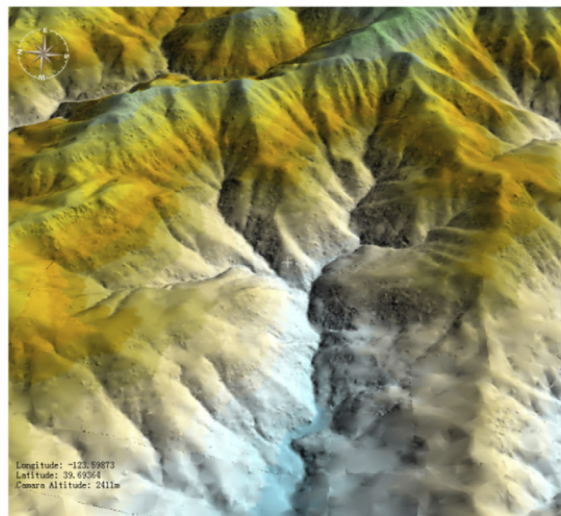
It should be pointed out that the proposed M-TIN method is actually more effective than expected in Table 2 in the comparison with the resampling method used in the traditional virtual globes. In this study, we used a grid mesh with a size of 256×256 vertices per tile for each level in the raster pyramid to maintain a resolution that matched the spatial scale of the texture image in this comparison. However, the recommended mesh size for elevation data is usually 17×17 or 33×33 vertices per tile in a typical grid-based virtual globe for real-time rendering, to reduce the transmission cost and guarantee the real-time frame rate. This results in a significant loss of accuracy for the generalized data in the raster pyramid, and thus the actual accuracy of the terrain data at different LODs in a grid-based virtual globe system could in fact be much lower than the results given in Table 2.



(a)



(b)



(c)

Fig. 17. Triangulation of large-scale LiDAR elevation points with the M-TIN approach for spherical rendering. (a) Rough LiDAR points. (b) Multi-resolution TIN wireframe. (c) Real-time terrain representation.

5.2. The improvement in terrain representation

Following the above discussions, a comparison between the terrain representation of the grid-based method and the TIN-based method is provided in Fig. 14. For the grid-based method, the raster elevation pyramid constructed by the bilinear resampling method was used as data support, and the mesh size was set as 33×33 vertices per tile. In Fig. 14(a), the large amount of geometric redundancy in the flat areas and the serious morphological distortion in the mountainous areas are clearly observable in the results of the grid-based terrain representation. Compared with the grid-based method, the proposed M-TIN method can achieve a more accurate and adaptive terrain representation, with some minor structures and the slope morphology of the landscape preserved. To allow a better visual inspection, the rendering results of the two models for the same scene are shown in Fig. 14(c). The grid-based terrain was rendered with the wireframe mode, while the TIN terrain was draped with image data. It can be seen that the grid-based method tends to raise up the depressions and cut down the protuberances, which results in an over-smoothed effect.

5.3. Morphological feature preservation

The accuracy of DEMs is often dependent upon how the features at different spatial scales are represented (Small and Sohn, 2015). It is therefore important to preserve the morphological nature of the landscape at the different scales. Fig. 15 provides a comparison of the terrain representation between the TIN structure with and without the drainage constraint. Clearly, the drainage features are effectively retained by enforcing the triangulation into the streamlines (red lines), as shown in Fig. 15(b). In previous studies, the linear patterns and the topological structure of the landscape have rarely been considered, leading to the absence of some significant morphological signatures for further applications. As can be seen in Fig. 15(a), the real-time simulation of the water flow movements can fail in the valley areas where the morphological information in the valleys has been lost in the multi-resolution modeling. It is therefore important to take the morphological structure information into account when constructing the multi-resolution TIN model.

5.4. Applications

Real-world terrain datasets are composed of different types of data with different topographical/morphological characteristics (Bóo et al., 2007). In the following two examples, we show that the proposed M-TIN method can be applied to different types of data and terrain and yield excellent results.

In the first example, the proposed M-TIN method was applied to a DEM from a coastal area of South Australia, which is characterized by high-contrast land-sea relief with a broken topological structure. Fig. 16 shows the real-time rendering effect of the land-sea topography based on the multi-resolution TIN pyramid generated by the proposed M-TIN method. The high visual quality and low geometric complexity are clearly visible from the TIN wireframe.

The rich morphological signatures provided by high-resolution LiDAR topography are essential for a deep understanding of Earth surface processes (e.g., landslide processes, geology, climate, and tectonic forcing processes) (Tarolli, 2014). However, most of the current virtual globe systems have difficulty in modeling and representing large-scale LiDAR elevation points. In this example, we applied the proposed M-TIN method to an arbitrarily distributed LiDAR elevation dataset from the area of the South Fork Eel River, California. The data are available online at the data distribution

archive (<http://www.opentopography.org>), and consist of roughly 8,000,000 elevation points, as shown in Fig. 17(a).

A multi-resolution TIN pyramid was first created for the LiDAR points, as shown in Fig. 17(a), and a final view-dependent LOD rendering was realized as the TIN wireframe shown in Fig. 17(b). A highly detailed terrain representation of the bare ground can be seen in Fig. 17(c). Since the proposed M-TIN method supports LOD triangulation and quadtree-based tiling, it permits the efficient processing and visualization of any large-scale LiDAR elevation dataset.

6. Conclusion

This study has presented a new method for the processing of large-scale elevation data for virtual globe visualization and analysis. Differing from the previous methods, the proposed approach takes cartographic principles into account when constructing the multi-resolution triangulated irregular network (TIN) pyramid. It connects each pyramid layer with a scale variable to formalize its LOD generation and enforce the topologically correct drainage features in the TIN structure. Hence, the morphological characteristics and the context of the landscape are preserved over a range of levels, while the data errors are well under the required data quality standard. In addition to the morphological LOD triangulation, the proposed approach also supports efficient tiling and storage of TINs with spatial structure, based on a critical virtual partitioning. Therefore, any regular or irregular elevation dataset can be used as input data, and an efficient out-of-core and view-dependent terrain rendering can be realized. In addition, an adaptive tessellation technique was developed to join TIN meshes with different LODs to eliminate run-time terrain cracks.

The use of the presented M-TIN method overcomes several of the limitations of the current virtual globe systems, and it can dramatically reduce data redundancy while achieving a morphologically preserved terrain representation. With the high-quality underlying terrain data and the high-fidelity terrain representation in the flexible TIN structure, virtual globes in the future could be extended beyond “geobrowsing” to include more scientific analysis and evaluation capabilities when handling global environmental change issues. The proposed M-TIN method was designed and implemented in our VirtualWorld v1.0 system. However, the core techniques used are universal and can be adapted to virtual globes for terrain processing and visualization. Growing demands for public participation make implementation of browser-based and VR/AR-based systems a hot issue in 3D visualization. Therefore, the extension of the proposed M-TIN method is a key point in the future work.

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